

The background of the slide is a photograph of the Modhera Sun Temple in Ahmedabad, India, during sunset. The temple's architecture, featuring a large circular staircase and a central temple with a dome, is reflected in a pool of water in the foreground. The sky is filled with soft, orange and yellow clouds.

IBM University Engagement — Project Submission

Smart Heritage Conservation Platform

Ahmedabad's Walled City & Modhera Sun Temple

Agentic AI with IBM Granite, IBM BOB & LangFlow | Domain: Tourism & Heritage

Project Details

Project Title: Smart Heritage Conservation Platform for Ahmedabad's Walled City & Modhera

Domain: Tourism & Heritage (Challenge 12)

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Problem Statement

The Challenge

Gujarat's UNESCO heritage sites — Ahmedabad's Walled City and the Modhera Sun Temple — face conservation risk from urban encroachment, weathering and unmanaged tourist footfall, while digital storytelling for visitors stays underutilised.

The Objective

- Monitor structural health of heritage structures continuously
- Manage and forecast visitor flow at peak hours
- Deliver immersive, personalised cultural storytelling



Proposed Solution

A multi-agent Agentic AI platform built on LangFlow, IBM BOB and IBM Granite LLM, hosted on IBM Cloud, that turns heritage conservation into a continuous, data-driven and visitor-facing service.

Structural Health Monitoring Agent

Analyses IoT sensor, drone and image data to detect cracks, erosion and moisture damage; predicts risk scores per structure.

Visitor Flow Management Agent

Forecasts footfall, recommends slot booking, alternate routes and crowd caps to reduce physical stress on monuments.

Personalized Heritage Storytelling Agent

Granite-powered RAG narratives in multiple languages, tailored to visitor interest, age and time available.

Encroachment Detection Agent

Compares satellite/CCTV imagery over time to flag illegal construction inside buffer zones.

Conservation Reporting Dashboard Agent

Auto-generates ASI/municipal-ready reports, priority repair lists and budget suggestions.

Technology Used

1 LangFlow

Visual orchestration of the multi-agent conservation workflow

2 IBM Granite LLM

granite-3-2-8b-instruct for reasoning, summarisation & storytelling

3 IBM BOB

Agent builder and orchestration layer for autonomous task execution

4 IBM Cloud

Secure, scalable hosting for agents, APIs and data pipelines

5 RAG + Vector DB

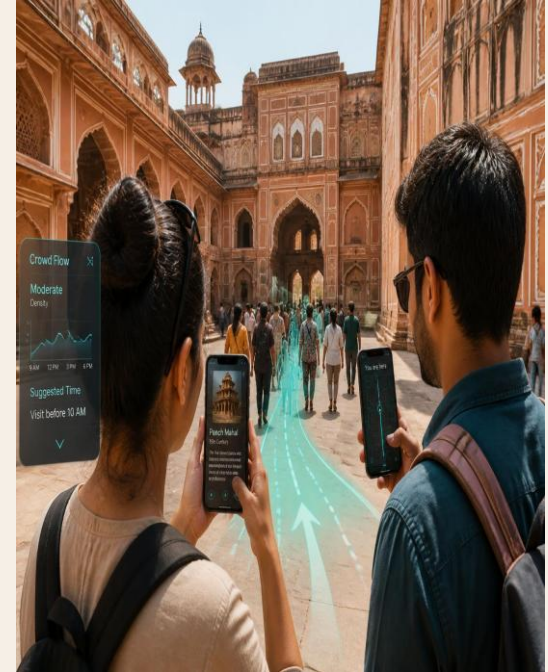
FAISS/Chroma over ASI records, heritage archives & inspection logs

6 IoT / Vision Inputs

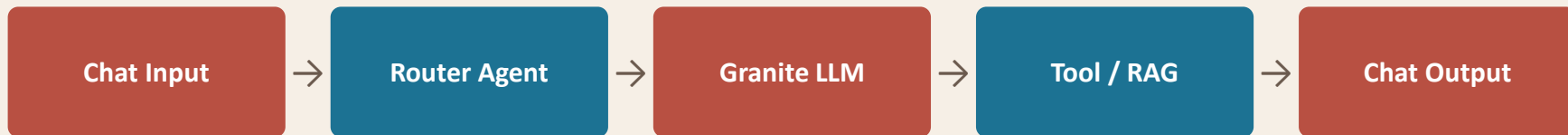
Crack sensors, drone imagery and CCTV feeds for site telemetry

LangFlow Components Used

- Chat Input — visitor or officer query (text / voice / image)
- IBM watsonx AI Agent — routes the request to the right agent
- IBM Granite Model — granite-3-2-8b-instruct for reasoning, narration and report drafting
- File / Data Component — ASI logs, sensor CSVs, site imagery
- Vector Store (Chroma) + Embeddings — RAG retrieval over heritage documentation
- Prompt Template — role prompts for each of the five agents
- Chat Output — narration, risk alert, crowd advisory or report

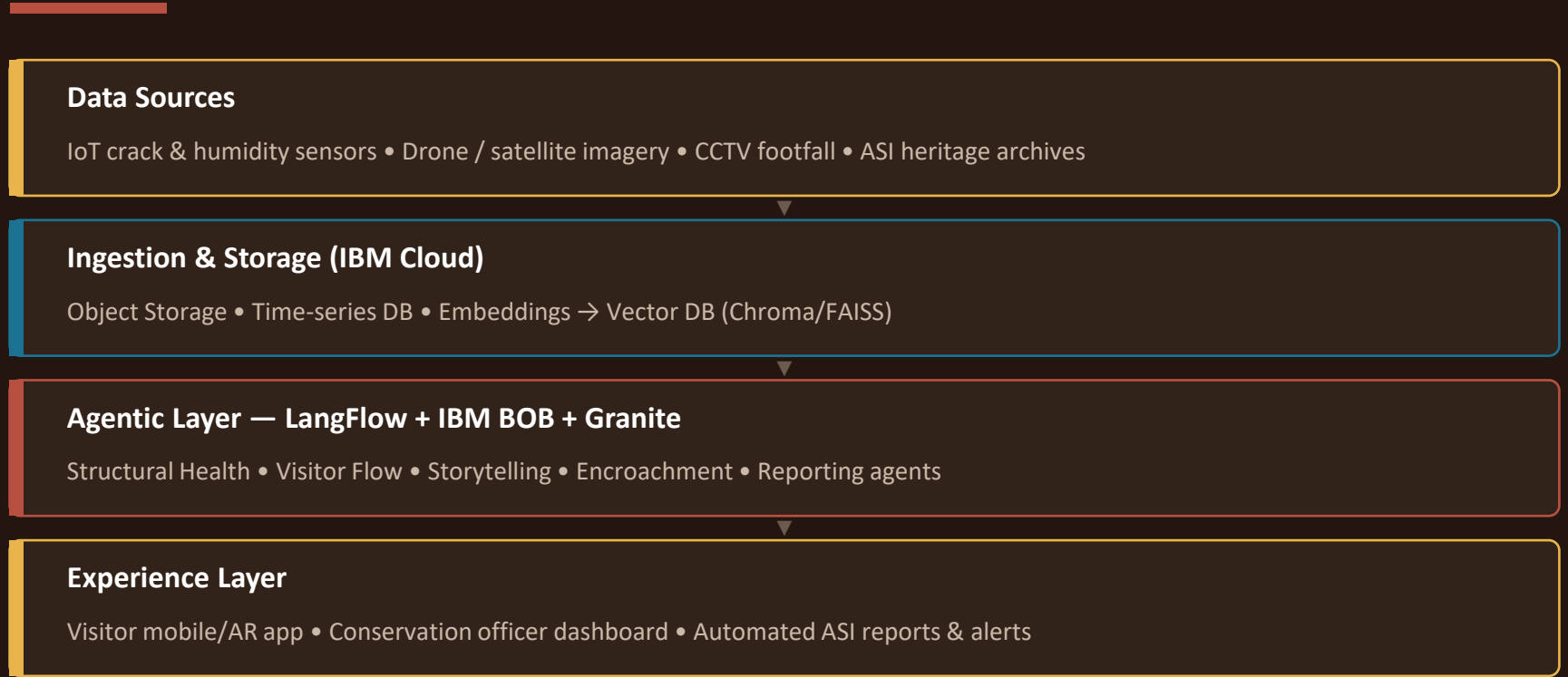


LangFlow Workflow



Flow: user query → agent router → Granite reasoning → RAG / sensor tools → grounded response back to visitor or conservation officer.

Architecture Blueprint



Data Sources

IoT crack & humidity sensors • Drone / satellite imagery • CCTV footfall • ASI heritage archives



Ingestion & Storage (IBM Cloud)

Object Storage • Time-series DB • Embeddings → Vector DB (Chroma/FAISS)



Agentic Layer — LangFlow + IBM BOB + Granite

Structural Health • Visitor Flow • Storytelling • Encroachment • Reporting agents

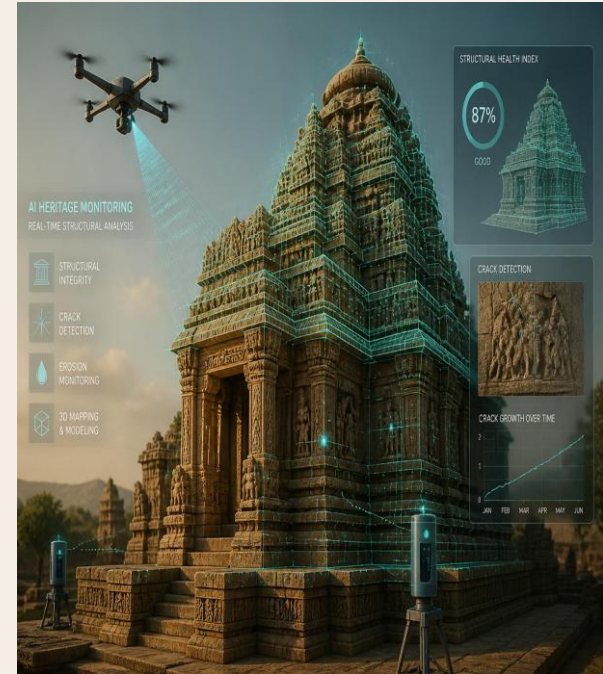


Experience Layer

Visitor mobile/AR app • Conservation officer dashboard • Automated ASI reports & alerts

Role of Agentic AI in the Solution

- Autonomy — agents run scheduled inspections and raise alerts without being prompted.
- Specialisation — five agents each own one conservation responsibility, then collaborate.
- Context awareness — decisions factor in monsoon season, festival crowds and monument age.
- Tool use — agents call sensor APIs, image models, ticketing systems and report generators.
- Continuous adaptation — risk scores and crowd advisories update in real time.
- Human-in-the-loop — every conservation action is proposed to ASI officers for approval.



Token Usage

Total tokens used: 116200

Task ×

1008 lines

Context Length 0.0k / 270.0k (0%) ⌵

Task Id 1a80a3d37d8a816e805b28c1290c84fb

Workspace playground

Bobcoins 💰 0.0

Before the Prompt

Task ×

is this project complete or not ?

Context Length 116.2k / 270.0k (43%) ⌵

Task Id 1a80a3d37d8a816e805b28c1290c84fb

Workspace playground

Bobcoins 💰 2.39

After the Prompt

Novelty and Uniqueness

1

Conservation + Tourism in one loop

Crowd data feeds structural risk; risk caps crowd — no existing platform closes this loop.

2

Predictive, not reactive

Granite forecasts deterioration before visible damage, prioritising repair budgets.

3

Hyper-local storytelling

Pol-level, multilingual narratives grounded in verified ASI archives via RAG.

4

Encroachment early warning

Time-series satellite diffing turns buffer-zone violations into same-week alerts.

5

Report automation

Officer-ready conservation reports generated in minutes instead of weeks.

6

Enterprise-grade & governed

IBM BOB + Granite on IBM Cloud with auditability and data governance.

GitHub Repository

Repo URL: <https://github.com/Meet-pyt/smart-heritage-platform.git>

Future Scope

AR / VR Time Travel

Reconstruct monuments to their original state and let visitors walk through the 15th-century Walled City in AR.

Digital Twin of Every Monument

A live 3D twin updated by drone scans, enabling simulation of monsoon and seismic stress.

Statewide Heritage Network

Extend from Ahmedabad and Modhera to Rani ki Vav, Champaner and Dholavira on one dashboard.

Citizen Reporting & Voice Access

Multilingual voice agent plus crowd-sourced damage reports from residents and guides.

Certificates Earned



Certificates Earned

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Meet Chauhan

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 01 Sep 2026 (GMT)

Learning hours: 30 mins



Thank You

Preserving Gujarat's heritage with Agentic AI — IBM Granite • IBM BOB • LangFlow • IBM Cloud